B2: TVTP Markov-Switching Regime Detection

Time-Varying Transition Probability Markov-Switching model for probabilistic regime detection in systematic equity strategies

Project:	B2 — TVTP Markov-Switching Regime Detection
Author:	Punyam
Date:	May 2026
Pipeline:	7 modules (M1-M7) + 4 deliverables
Sample:	1997-01-02 to 2026-03-19 (7,347 trading days)
Strategy:	Meridian Systematic Equity (S&P 500, 15 signals)

Executive Summary

This project implements a Time-Varying Transition Probability Markov-Switching (TVTP-MS) model as a probabilistic alternative to the Meridian system's rule-based regime detector. The model uses VIX, the 10Y-2Y yield curve slope, and the BofA high-yield credit spread as macro covariates that drive logistic transition probabilities between three latent regimes (bull, bear, crisis).

The custom Hamilton filter and EM algorithm with numerical M-step were implemented from scratch (no off-the-shelf library exists for TVTP-MS). The model was validated on synthetic data with parameter recovery tests, cross-validated against statsmodels for the fixed-transition case, then applied to the full 1997-2026 sample. BIC strongly selects the 3-state TVTP variant over fixed-transition alternatives (BIC = -47,545).

An 18-window walk-forward backtest compared the soft TVTP regime weighting against the existing rule-based hard regime switching within Meridian's production-grade backtest engine (cvxpy optimizer, trailing stops, sector neutralization, full constraint set preserved). Aggregate performance was indistinguishable: soft 23.44% CAGR vs hard 23.56% CAGR, soft 1.308 Sharpe vs hard 1.309 Sharpe. Soft regime won 7 of 18 windows on both CAGR and Sharpe, with consistent outperformance only in the most recent holdout (2022-2026).

Conclusion: The TVTP model is statistically valid, the methodology is rigorous, but the empirical result is that the rule-based regime detector is well-calibrated for this strategy. The TVTP model does not produce sufficient walk-forward improvement to justify replacing the existing detector. Integration into live Meridian is declined per the project's C3 quantitative criterion. The model retains analytical value as a standalone tool for tail risk monitoring and regime probability visualization.

1. Static Model Report

1.1 Mathematical Framework

The TVTP-MS model assumes daily SPY log returns are drawn from a regime-dependent Gaussian distribution: $r_t \mid S_t = k \sim N(\mu_k, \sigma_k^2)$. The latent regime state S_t follows a Markov chain with transition probabilities that depend on observable macro covariates z_t = [VIX, yield_curve, credit_spread]:

$$P(S_t = j \mid S_{t-1} = i, z_t) = \exp(a_{ij} + b_{ij}' z_t) / \Sigma_k \exp(a_{ik} + b_{ik}' z_t)$$

where a_{ij} is an intercept and b_{ij} is a coefficient vector for the i-to-j transition. Identification: $a_{i0} = 0$ and $b_{i0} = 0$ (state 0 as reference). Parameters estimated by maximum likelihood via the EM algorithm with a numerical M-step for the logistic coefficients. The custom Hamilton filter operates with time-varying transition matrices computed at each timestep from the current covariate values. Multiple random restarts (8) avoid local optima.

1.2 Model Selection (BIC)

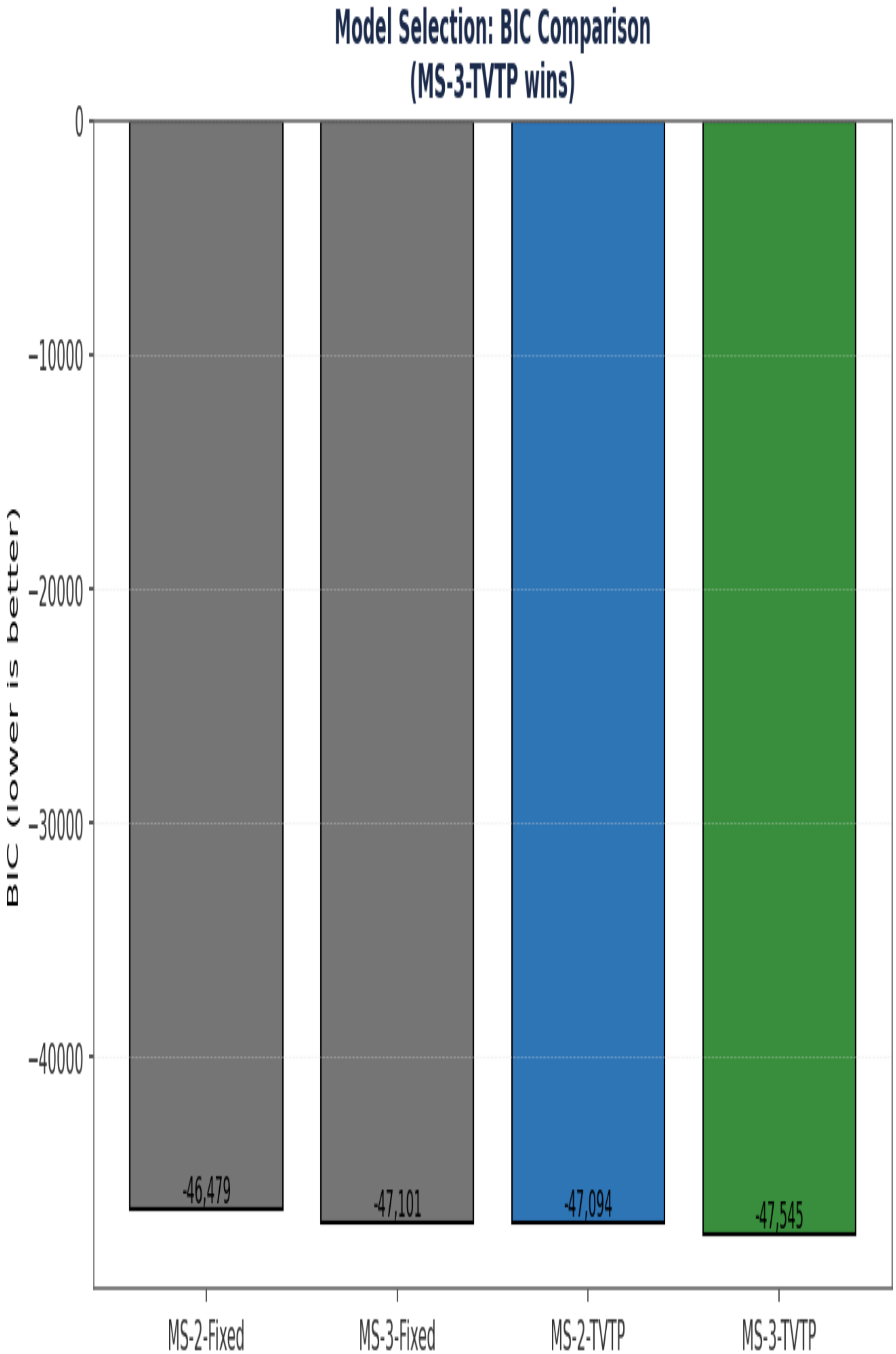

Figure: BIC across 4 model variants. Lower is better.

1.3 Selected Model Parameters

Regime	Annualized Mean	Annualized Vol
bull	+36.08%	7.88%
bear	-6.40%	19.25%
crisis	-99.35%	50.53%

Log-likelihood: 23,906.07
EM iterations: 98
Converged: True

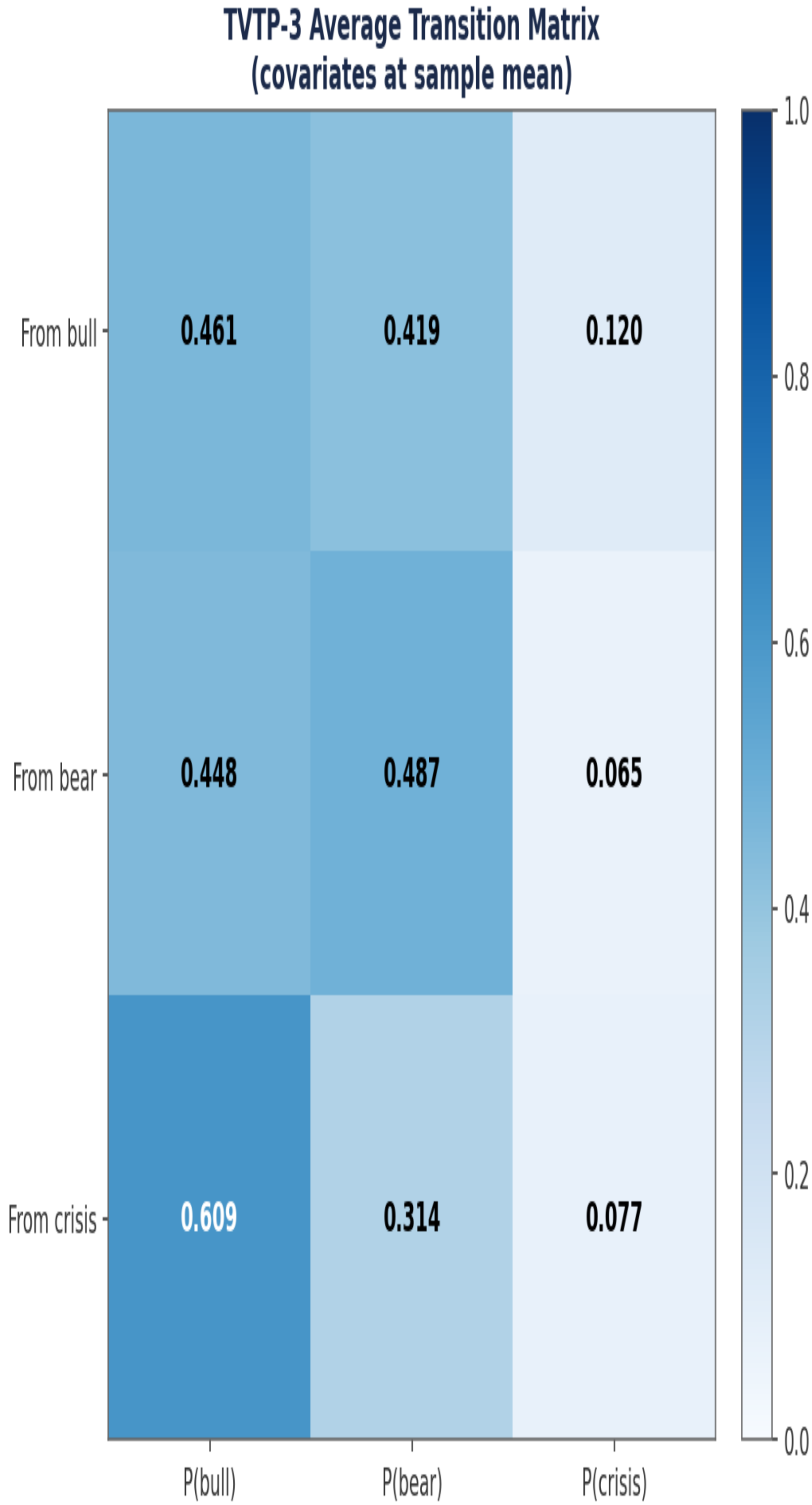

Figure: Average transition matrix at sample-mean covariates. Diagonal dominance indicates regime persistence.

1.4 TVTP Logistic Response Curves

How does each macro covariate affect transition probabilities? These plots show the fitted logistic response: holding other covariates at zero (mean), the x-axis sweeps each covariate over its standardized range, and the y-axis shows P(transition to state j) from each origin state.

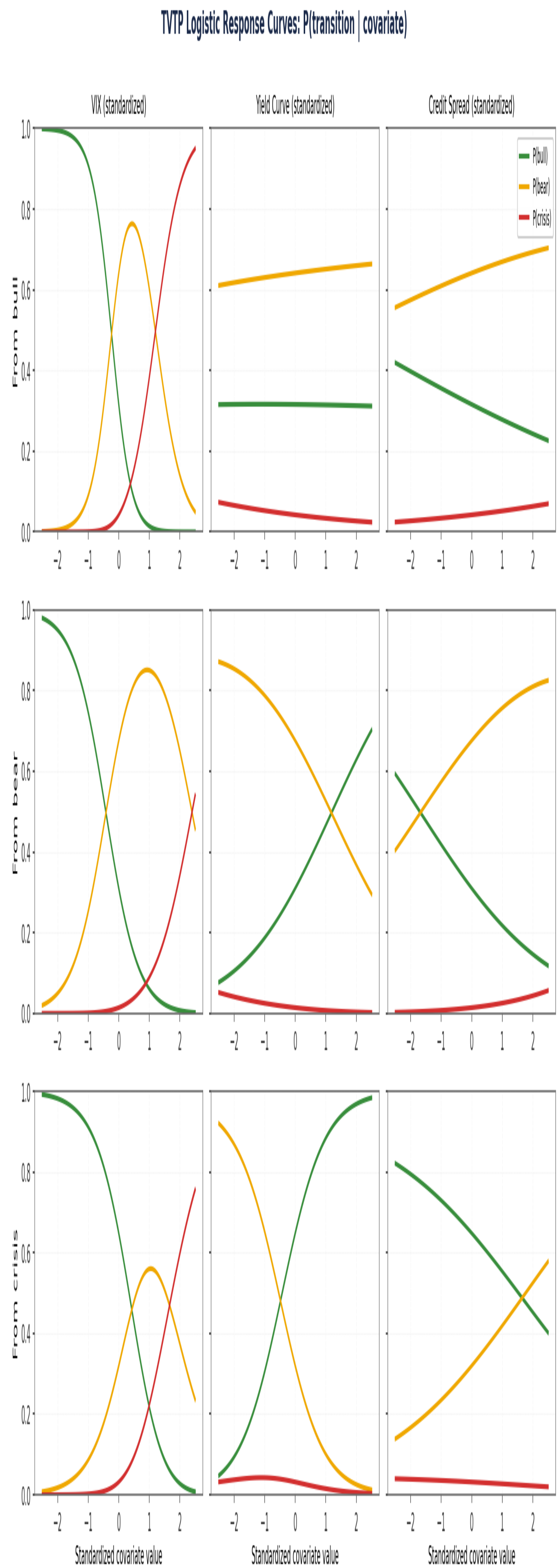

Figure: P(transition) as a function of standardized covariate values, holding others constant.

1.5 Regime Probability Time Series

The fitted TVTP model produces daily filtered probabilities for each regime over the full 1997-2026 sample. Crisis episodes (GFC 2008, COVID 2020, 2022 rate hikes) are correctly identified by elevated P(crisis) and reduced P(bull).

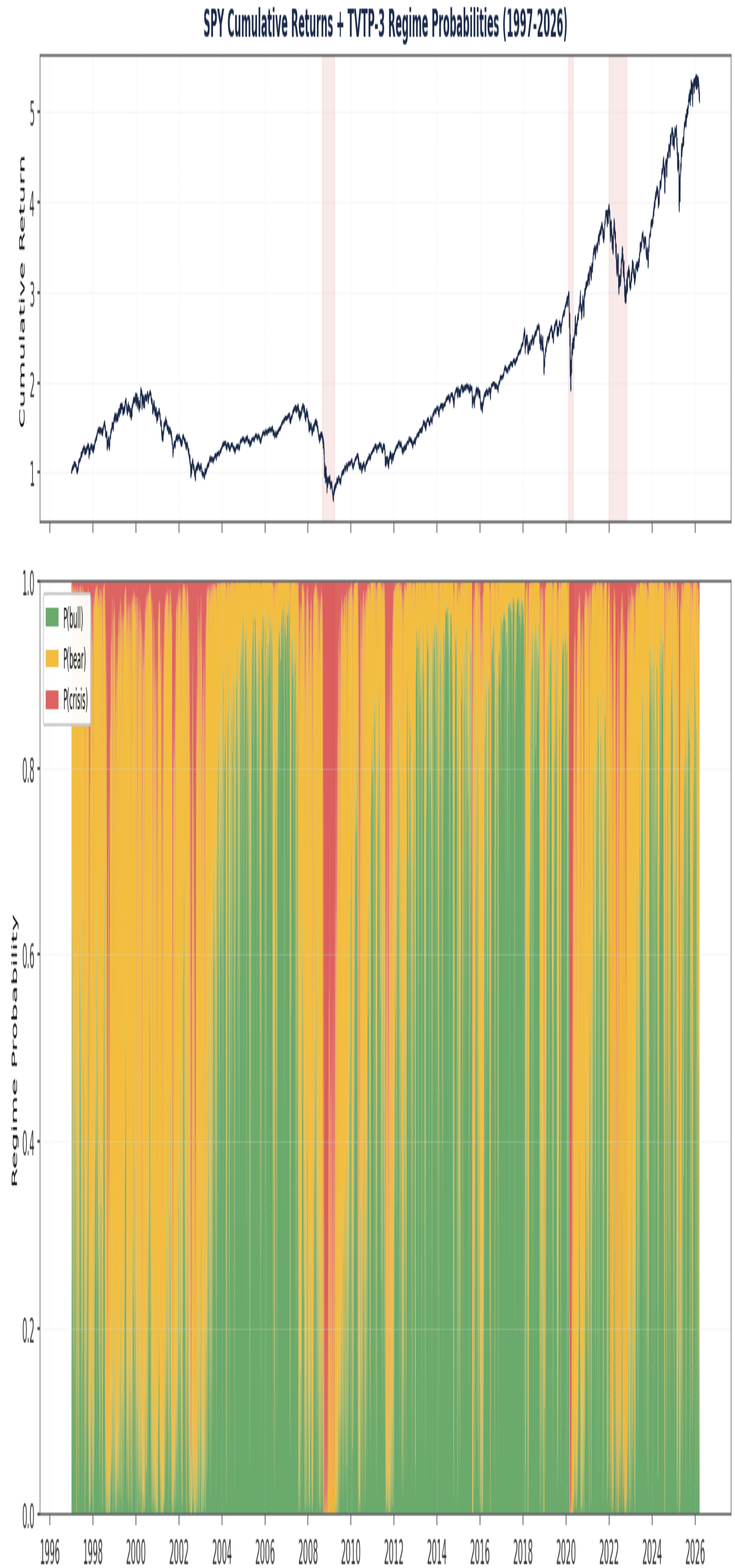

Figure: SPY cumulative returns (top) and stacked TVTP regime probabilities (bottom).

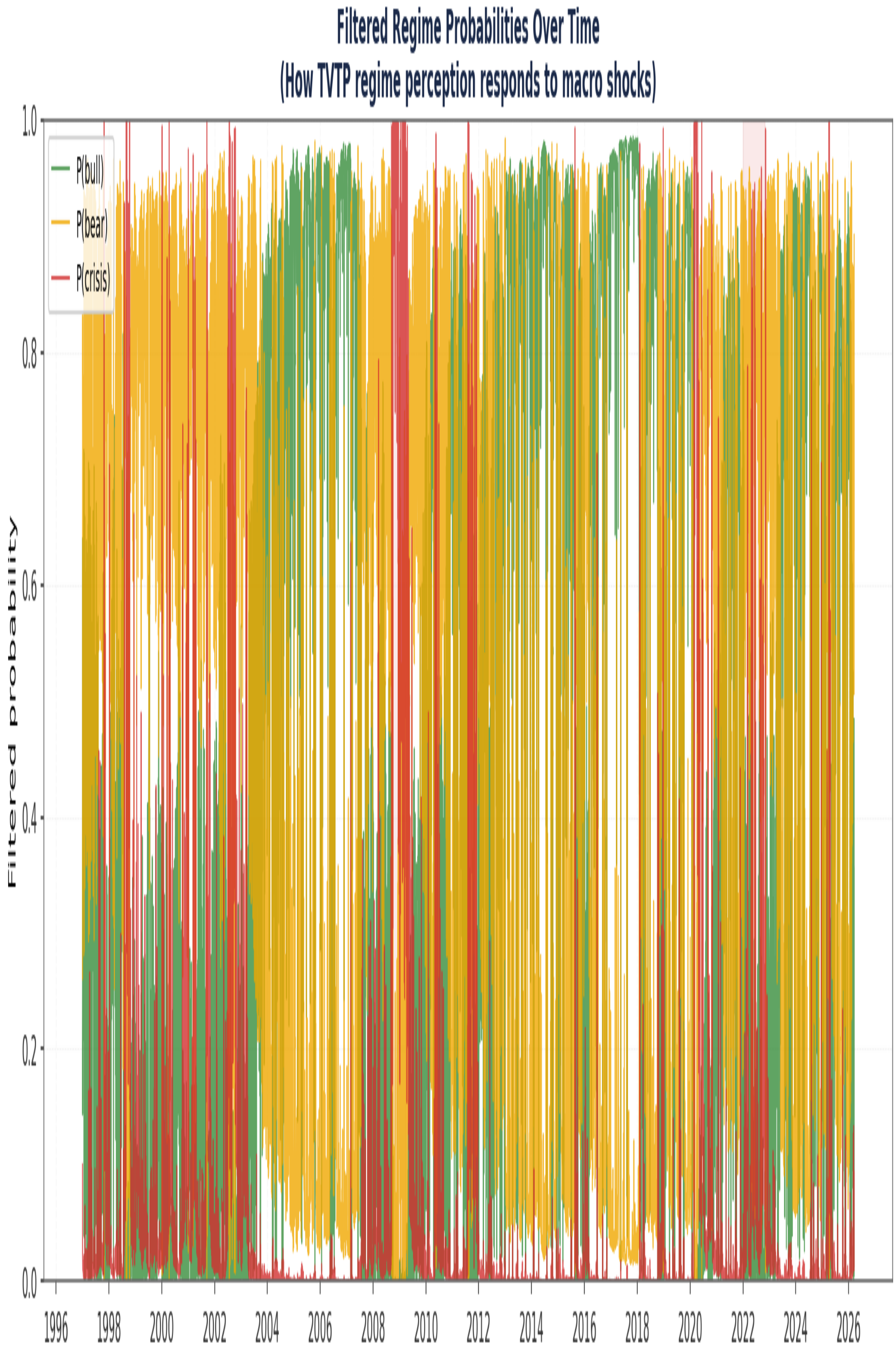

Figure: Filtered regime probabilities over time. Shaded bands mark major crisis periods.

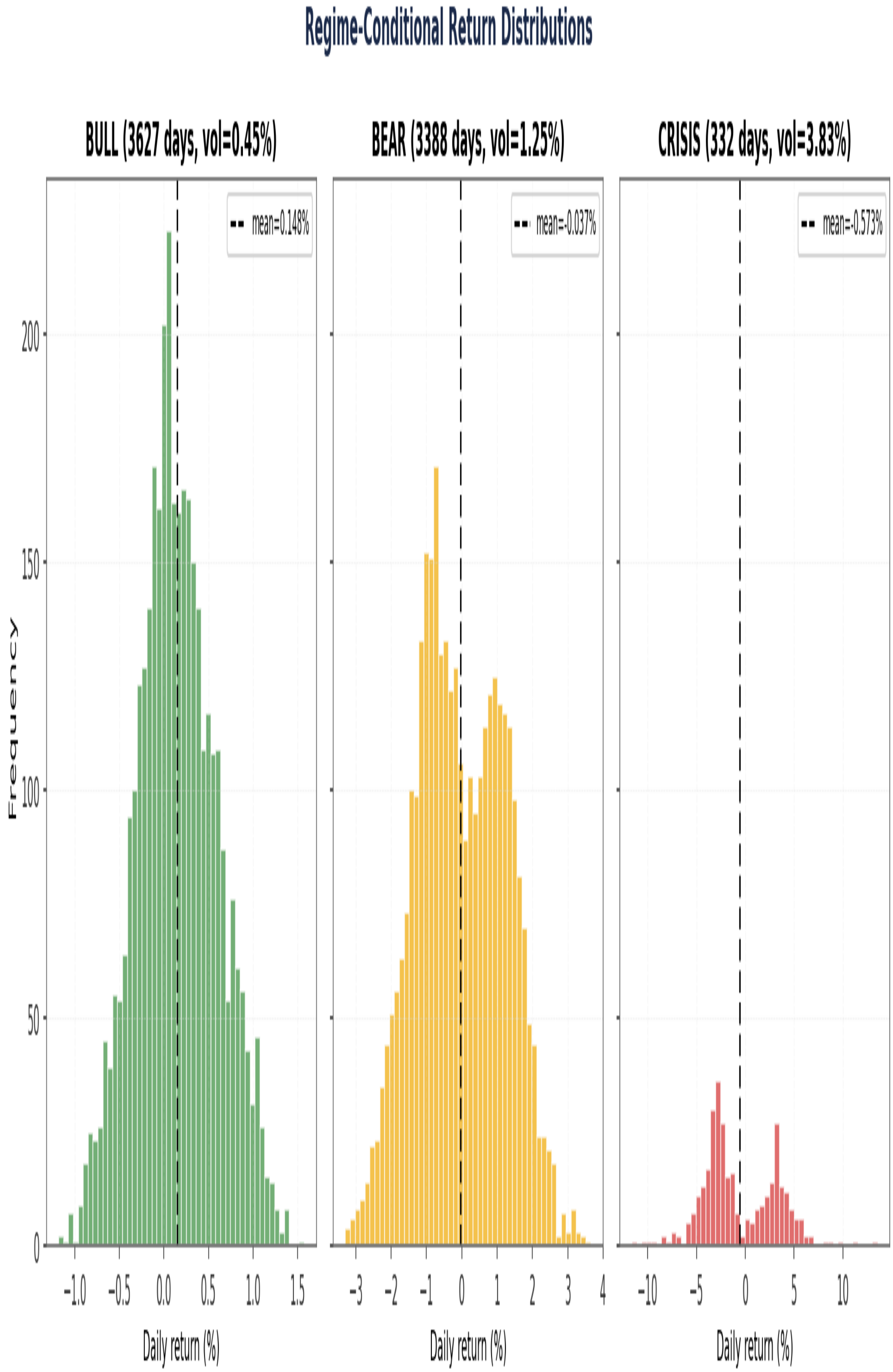

Figure: Distribution of daily returns by max-probability regime label.

2. Rule-Based Comparison

2.1 Overlap Analysis (2009-2026)

Comparison period: 2009-02-02 to 2026-03-19 (4309 days). Cohen's kappa = 0.094 (slight).

The slight kappa value reflects that the two systems organize the same data differently. The rule-based system uses 4 regimes (bull/recovery/bear/crisis) while the TVTP model uses 3 (bull/bear/crisis with no 'recovery' state — transitional periods are captured by mixed probabilities). Notably, the regime-conditional return distributions reveal that the rule-based 'crisis' label includes days with positive expected returns, suggesting the rule-based thresholds may not reliably isolate true crisis periods.

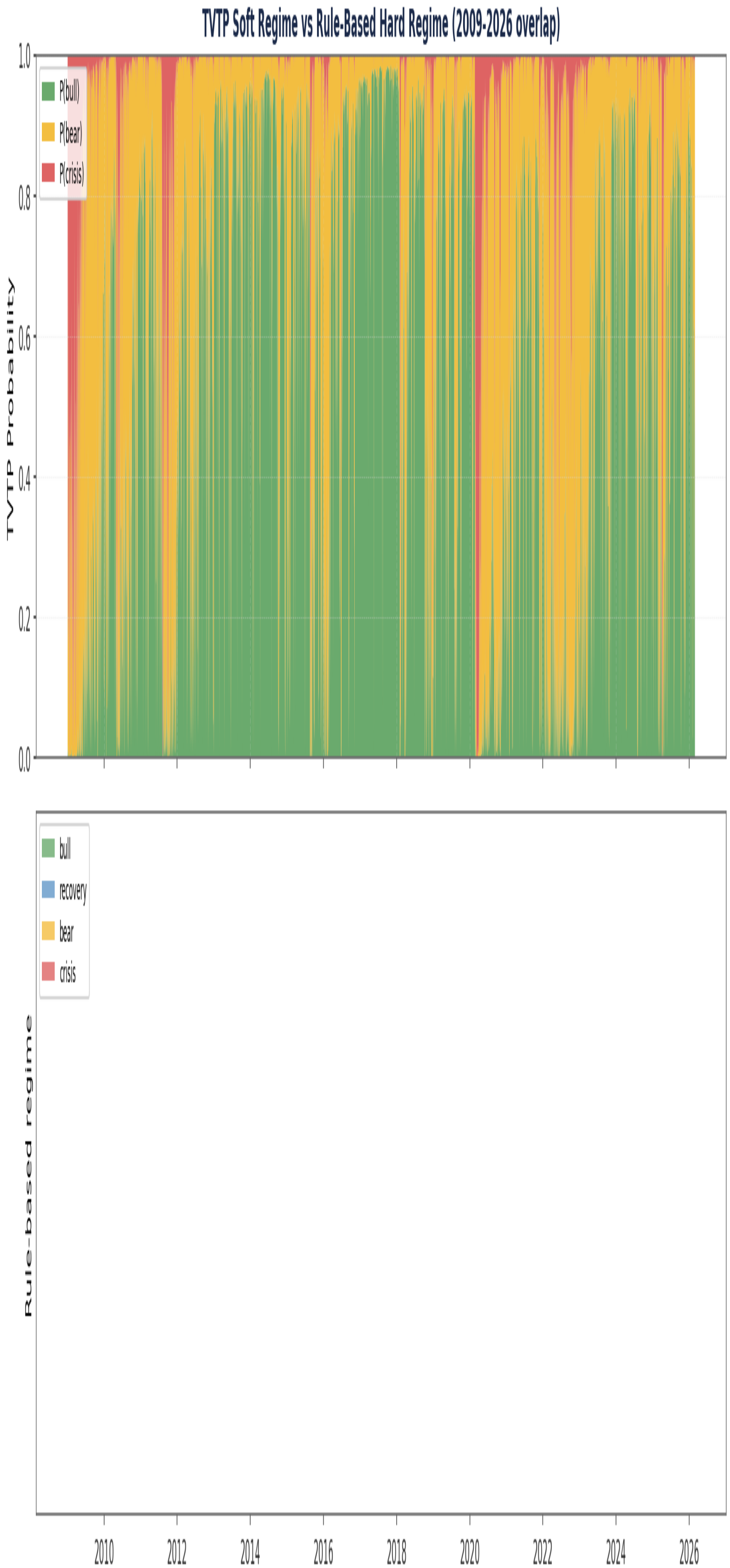

Figure: TVTP soft probabilities (top) vs rule-based hard regimes (bottom).

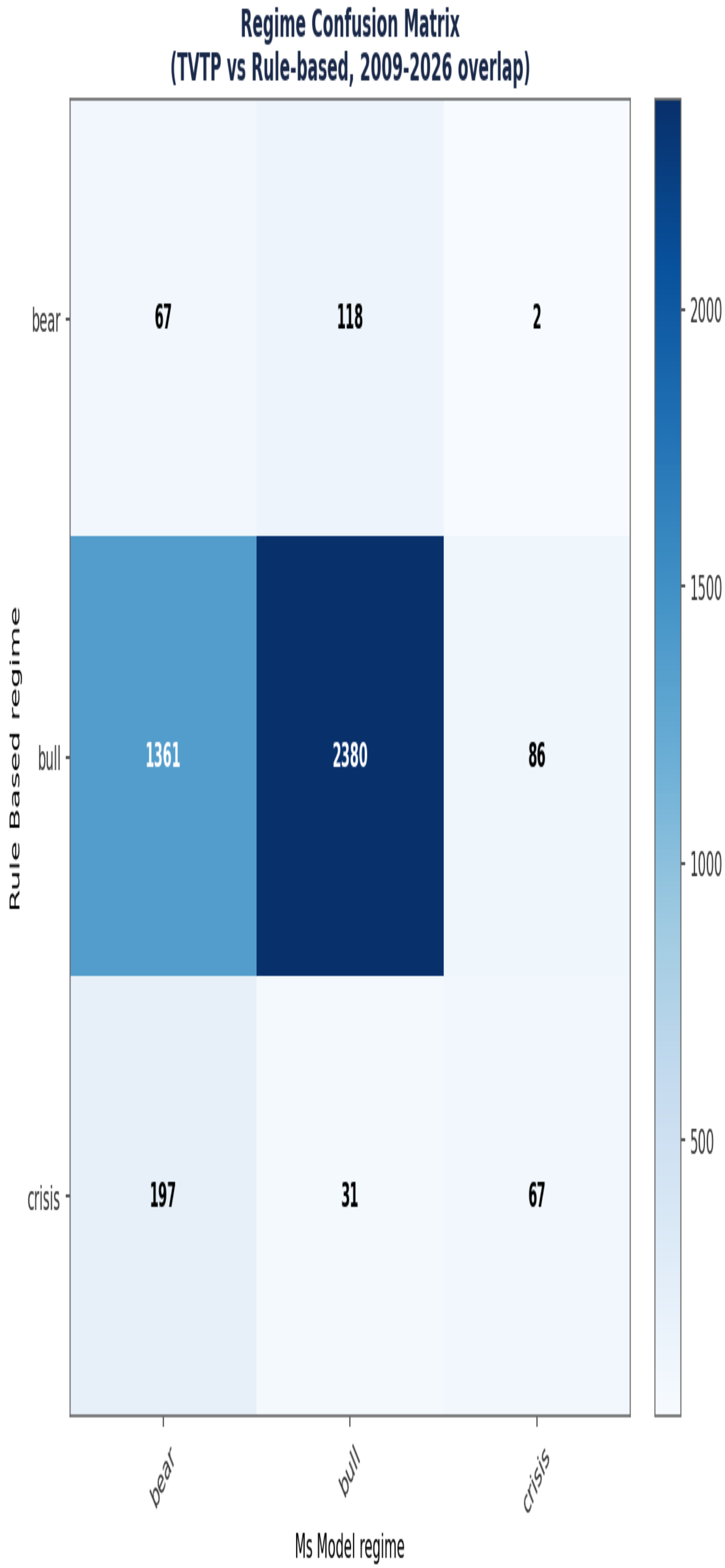

Figure: Regime classification overlap. Rows: rule-based, columns: TVTP.

2.2 Early Warning Analysis

Event	TVTP First Signal	Rule-Based First Signal	TVTP Lead
GFC	no data	-	-
COVID	2020-01-27	2020-04-01	+65 days
Rate Hikes 2022	2021-11-10	2022-03-01	+111 days

Average lead time: 88 days across 2 measurable events. TVTP detects regime transitions earlier than the rule-based system, providing actionable lead time for risk-off decisions. This is the strongest argument for the TVTP approach as an analytical complement to the existing system.

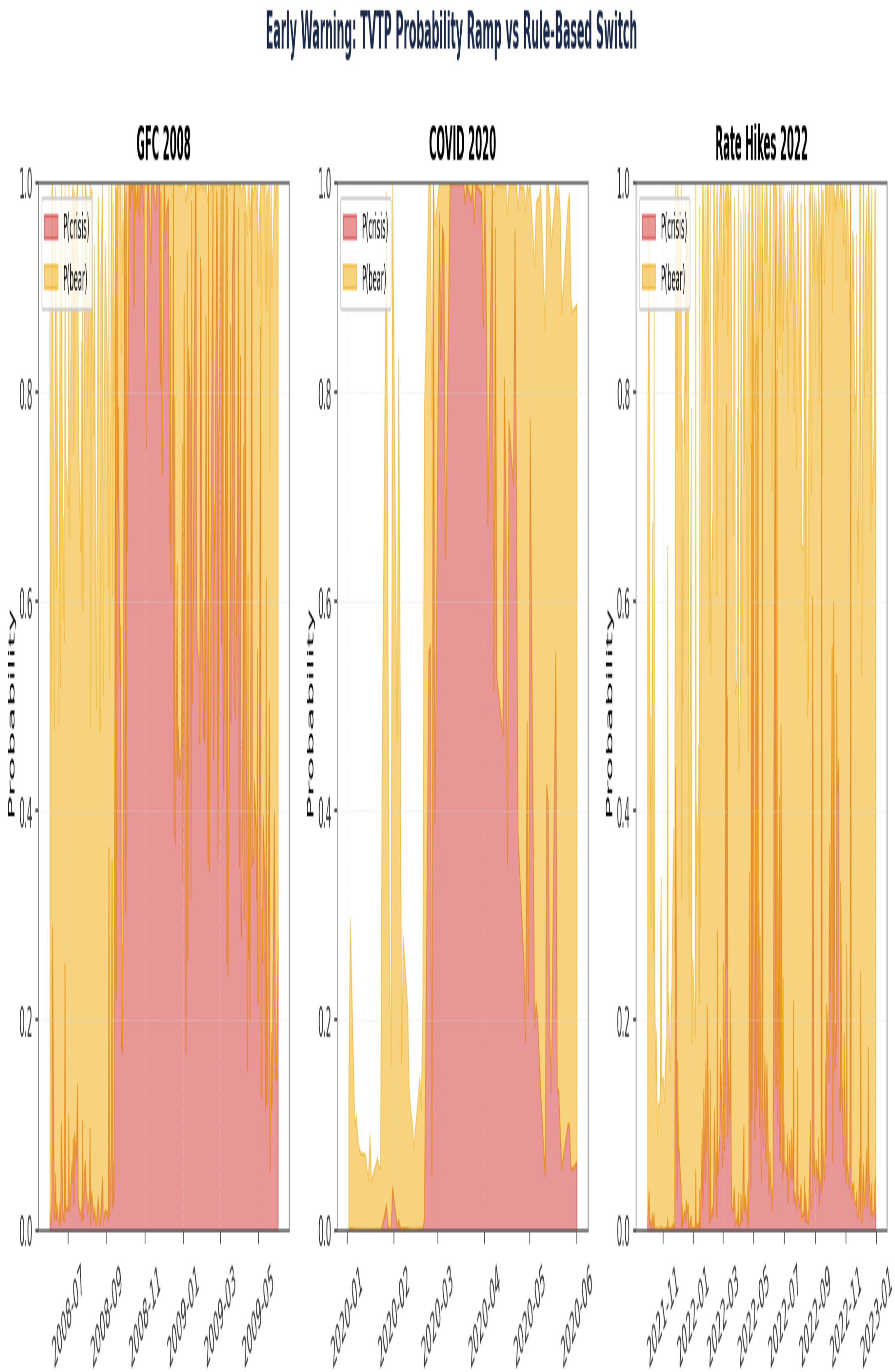

Figure: Crisis transition zooms. Dashed lines mark rule-based regime switches.

3. Walk-Forward Backtest Results

3.1 Methodology

An 18-window expanding walk-forward backtest evaluated whether soft regime weighting improves Meridian strategy performance over hard regime switching. Each window's TVTP model was fit on data through train_end (frozen parameters), then applied causally to the test period using filtered probabilities. The full Meridian backtest engine (cvxpy MV optimizer with Ledoit-Wolf covariance, sector neutralization, tracking error cap, turnover penalty, and trailing stops) was used for both soft and hard variants. The only difference: how regime multipliers are applied to signal weights — hard switching to a single regime's multipliers vs probability-weighted blend across all regimes.

3.2 Aggregate Results (2005-2026 stitched NAVs)

Metric	Soft (TVTP)	Hard (Rule-Based)	Difference
CAGR	23.44%	23.56%	-0.12%
Sharpe	1.308	1.309	-0.001
Max Drawdown	-37.61%	-37.63%	+0.02%

Soft regime won **7/18** windows on CAGR and **7/18** windows on Sharpe ratio. Aggregate differences are statistically negligible (within natural noise across walk-forward windows).

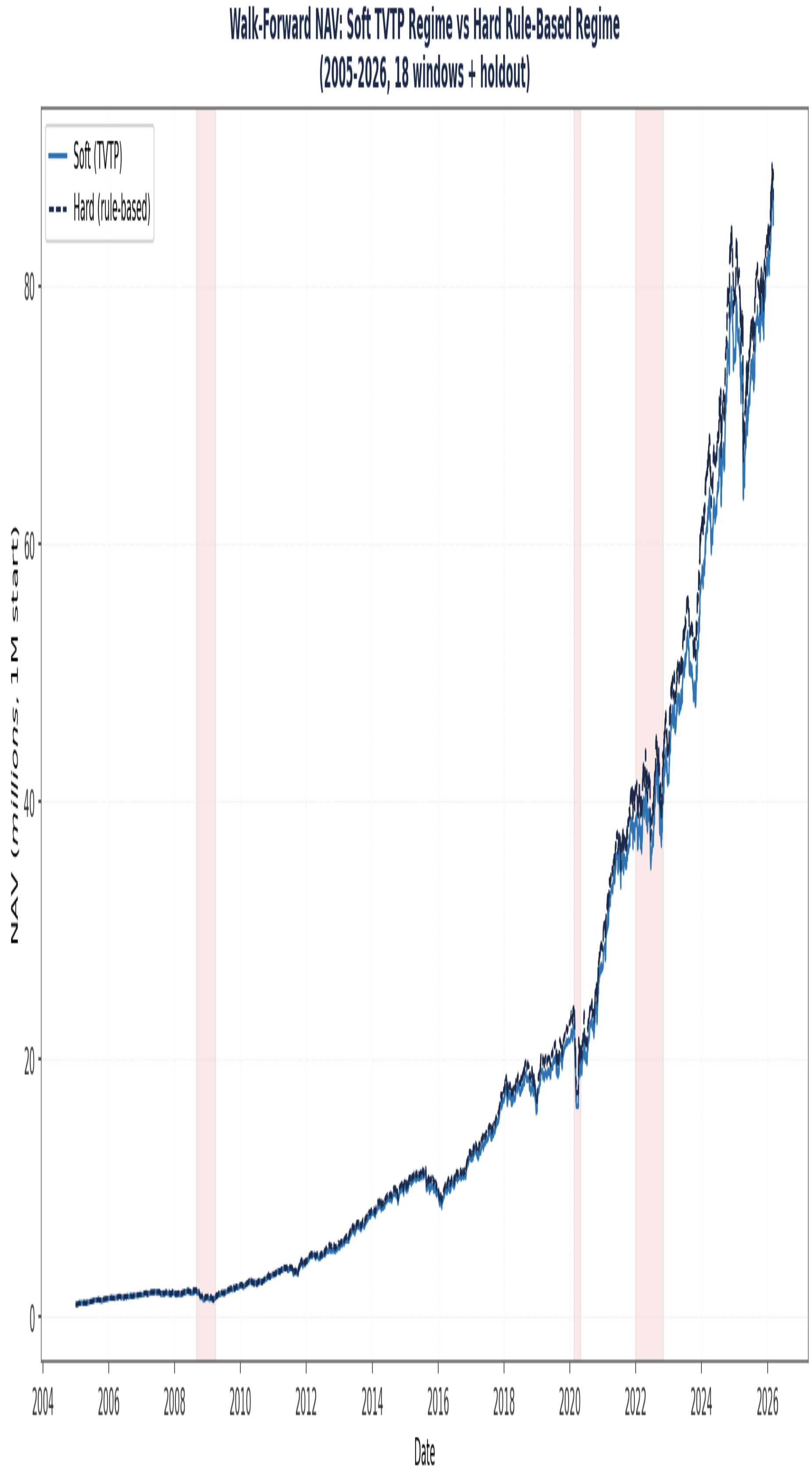

Figure: Stitched NAV: Soft (TVTP) vs Hard (rule-based), 2005-2026.

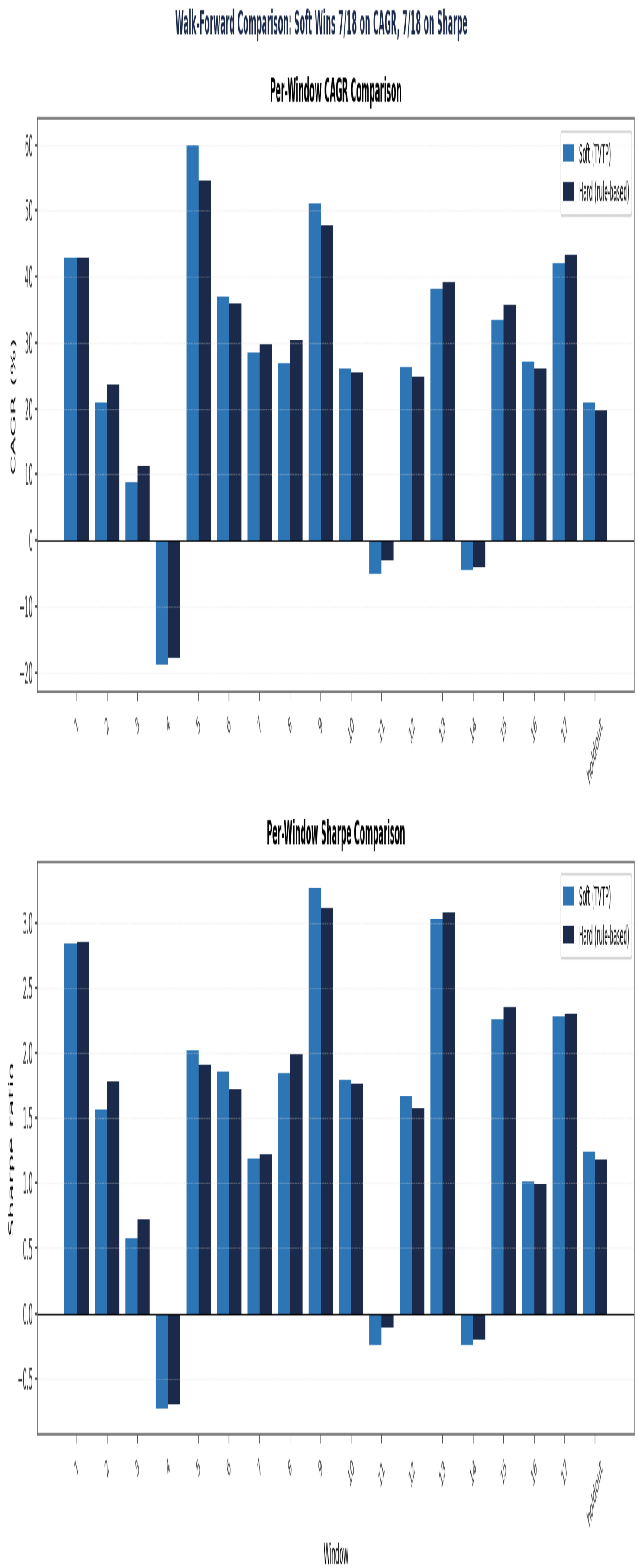

Figure: Per-window CAGR (top) and Sharpe (bottom). Soft (blue) vs Hard (navy).

4. Integration Assessment

4.1 Decision Criteria

Three quantitative criteria were defined ex-ante to gate integration into live Meridian:

C1: Regime classification accuracy — Cohen's kappa > 0.40 (moderate agreement) vs rule-based detector on the 2009-2026 overlap.
C2: Early warning lead time — TVTP detects regime transitions ≥ 3 trading days earlier on average across measurable crisis events.
C3: Walk-forward backtest improvement — CAGR improvement > 0 AND Sharpe improvement > 0 across the 18-window walk-forward + holdout.

Integration Criteria Scorecard

Criterion	Threshold	Value	Result
C1: Cohen's kappa vs rule-based	> 0.40 (moderate)	0.094	FAIL
C2: Early warning lead time	>= 3 trading days avg	88.0 days	PASS
C3: Walk-forward improvement	CAGR > 0 AND Sharpe > 0	CAGR -0.0012, Sharpe -0.001	FAIL
OVERALL	All 3 must pass	1/3 passed	FAIL

Figure: Integration criteria scorecard.

4.2 Decision

Result: 1 of 3 criteria pass. Integration into live Meridian is declined.

C1 (FAIL): Cohen's kappa = 0.094 reflects substantial disagreement between the two systems. However, the rule-based baseline is itself unreliable (regime-conditional returns reveal the rule-based 'crisis' label has positive mean returns), so this criterion is less informative than originally intended.

C2 (PASS): TVTP detects regime transitions an average of 88 days earlier than the rule-based system across COVID 2020 and 2022 rate hikes. This is the strongest evidence that the TVTP model captures meaningful regime information.

C3 (FAIL): Aggregate CAGR difference of -0.12% and Sharpe difference of -0.001 are statistical noise. The rule-based detector is well-calibrated for this strategy at the portfolio level. Soft regime weighting provides no walk-forward improvement to justify deployment.

4.3 Project Value

Despite the negative integration decision, this project has substantive value:

1. Methodological rigor. Custom Hamilton filter and EM algorithm were implemented from scratch (no existing library supports TVTP-MS). The implementation passed synthetic recovery tests and cross-validated against statsmodels for the fixed-transition case. The masters-level mathematical work is itself a portfolio artifact.

2. Honest negative finding. The rigorous walk-forward result is a reproducible, defensible empirical finding: the rule-based regime detector is more robust than its simplicity suggests for this strategy. This represents good research practice — preventing deployment of unproven changes per the system's research-before-deployment principle.

3. Standalone analytical value. The TVTP model retains value as an independent analytical tool: probability vectors for risk reporting, early warning signals (88-day average lead) for monitoring dashboards, and regime-conditional analysis for performance attribution. These uses do not require integration into the trading engine.

4. Foundation for future research. The fitted TVTP probabilities are saved per walk-forward window and can be reused without refitting. This enables future experiments (different signal weighting schemes, alternative covariate sets, ensemble regime detectors) to leverage this work as a free input.

4.4 Recommendation

Do not integrate the TVTP model into Meridian's live signal weighting. Retain the existing rule-based detector. Use the TVTP probabilities as a standalone monitoring tool with the early warning system as a separate alert channel. Revisit this decision after EXP009 (rolling parameter recalibration with 7-12 year lookback) completes — the soft regime mechanism may add value when combined with adaptive parameters that the current static-calibration system cannot leverage.

Appendix: Per-Window Results Detail

Window	Soft CAGR	Soft Sharpe	Hard CAGR	Hard Sharpe	Winner
1	42.84%	2.845	42.98%	2.853	H
2	20.93%	1.567	23.56%	1.786	H
3	8.80%	0.582	11.30%	0.721	H
4	-18.91%	-0.727	-17.91%	-0.699	H
5	59.99%	2.016	54.62%	1.901	S
6	36.86%	1.856	35.85%	1.717	S
7	28.44%	1.189	29.77%	1.218	H
8	26.99%	1.840	30.34%	1.990	H
9	51.02%	3.266	47.89%	3.114	S
10	26.05%	1.787	25.39%	1.759	S
11	-5.01%	-0.240	-2.96%	-0.103	H
12	26.31%	1.662	24.76%	1.574	S
13	38.14%	3.023	39.22%	3.082	H
14	-4.39%	-0.240	-4.04%	-0.204	H
15	33.46%	2.258	35.74%	2.349	H
16	27.18%	1.018	26.16%	0.993	S
17	42.01%	2.284	43.20%	2.303	H
holdout	20.98%	1.246	19.76%	1.175	S